Two-joint Robotic Arm Simulation

Task 2 | Model, motion and reproducible simulation evidence

Model and degrees of freedom

The manipulator has two revolute joints (2R): shoulder angle q1 and elbow angle q2. These are two independent positioning DOF in the x-y plane. A binary ideal gripper is modelled separately; it is not a third positioning axis. Tool orientation is not independently controlled. Link lengths are L1 = 250 mm and L2 = 180 mm. Coordinates use the base as (0,0), x right, y up. This is a 2D kinematic simulation, permitted by the brief; it is not a SolidWorks assembly or a physically qualified robot.

Forward and inverse kinematics

$x = L1 \cos(q1) + L2 \cos(q1+q2)$; $y = L1 \sin(q1) + L2 \sin(q1+q2)$. Define $c2 = (x^2+y^2-L1^2-L2^2)/(2\ L1\ L2)$. With the selected positive-elbow branch, $q2 = \text{acos}(c2)$; $q1 = \text{atan2}(y,x) - \text{atan2}(L2 \sin(q2), L1+L2 \cos(q2))$. Targets outside $|c2| <= 1$ are rejected. The ideal radial workspace is 70 to 430 mm, before joint limits. Configured limits: q1 from -100 to 180 degrees, q2 from 5 to 170 degrees. These limits are design assumptions, not manufacturer specifications.

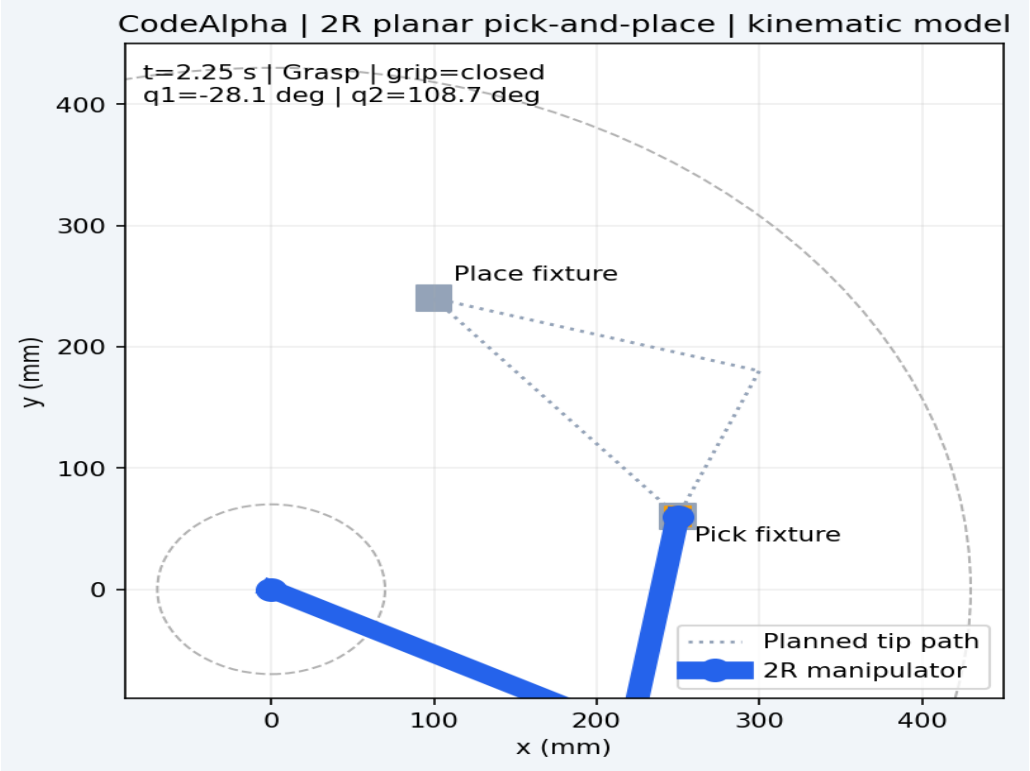
Motion sequence

HOME (300,180) -> PICK (250,60), 2 s; grasp dwell, 0.5 s; transfer to PLACE (100,240), 3 s; release dwell, 0.5 s; return home, 2 s. Each motion segment uses Cartesian interpolation $u = 3s^2 - 2s^3$, with zero endpoint velocity. Every sampled point is checked by inverse kinematics. While held, the object follows the tool; after release, it remains at PLACE. The ideal fixtures indicate locations and are not solid collision geometry.

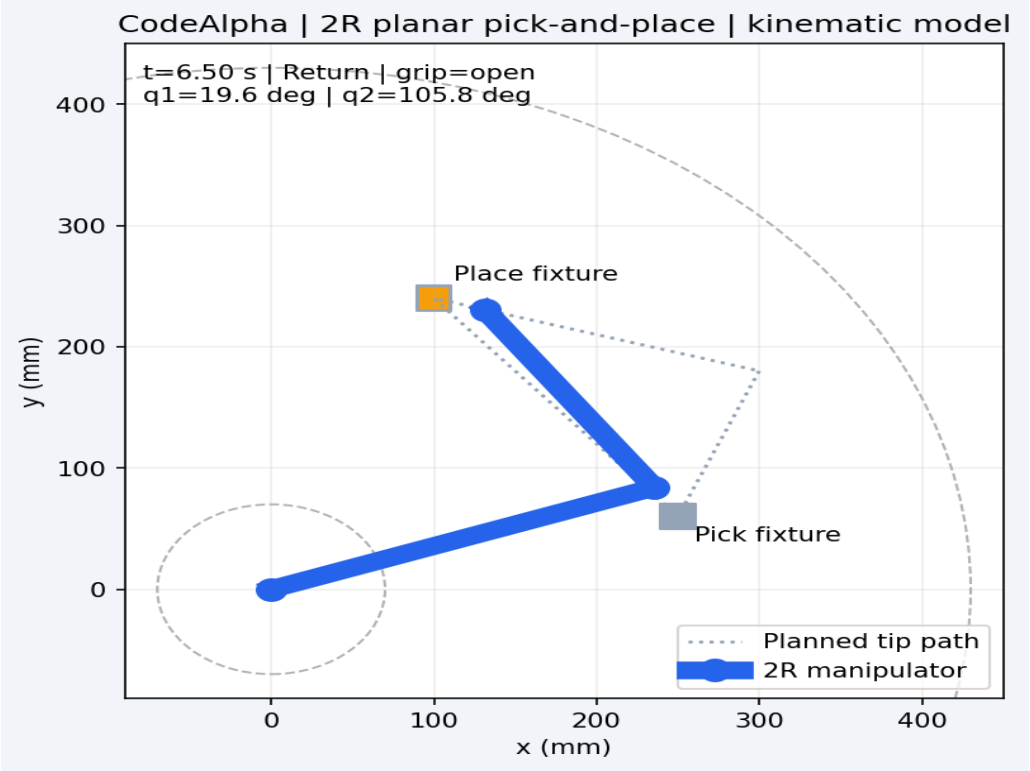
Run and inspect

No-install option: open arm_demo.html in a browser and press Play. Reproducible Python option: install the packages in requirements.txt, then run python arm_simulation.py from this folder. It regenerates the GIF, four images, trajectory CSV/JSON and result JSON. Run python test_arm.py for the four verification tests. Numerical data are precomputed in the browser viewer from the same Python trajectory.

Simulation frames



Grasp: object attaches at the pick fixture.



Release completed: object remains at the place fixture as the arm returns.

Results and limits

Metric	Measured simulation output
Duration	8.0 s
Samples	161
Maximum FK residual	8.039e-14 mm
Final object residual	0.000e+00 mm
q1 range	[-28.06, 26.1] deg
q2 range	[72.21, 117.66] deg

What was verified

Four automated tests passed: forward/inverse kinematics agree at every sample; unreachable targets are rejected; a held object follows the tool and is released at PLACE; consecutive tool positions remain continuous within the stated sample-step threshold. The complete 8-second motion was rendered into a GIF and four static simulation frames.

What these results mean

The very small residual is floating-point agreement within an ideal model. It is not measured physical robot accuracy. The project excludes inertia, torque, backlash, friction, actuator bandwidth, object contact and collision detection. A physical build would need these analyses, payload and motor sizing, tooling design and application safety assessment. No speed or payload capability is claimed.

Software references

[10] Matplotlib contributors. *Animation API documentation*. n.d.. Accessed 7 October 2026. https://matplotlib.org/stable/api/animation_api.html

[11] Python Software Foundation. *unittest: Unit testing framework*. n.d.. Accessed 7 October 2026. https://docs.python.org/3/library/unittest.html